# Ayush Nambiar

## Robotics & Control Engineer

RESUME_1918

Location: Gurugram, Haryana, India | Email: ayush.nambiar.resume_1918@example.com | Phone: +91-00000-00000 (placeholder)
Total Experience: 2 years

## PROFESSIONAL SUMMARY

Robotics & Control Engineer with 2 years of experience in robotics, using ROS 2, Python, C++, MATLAB. Experienced in autonomous navigation, control loops and measurable delivery outcomes.

## TECHNICAL SKILLS

Skills: C++, SLAM, Simulink, OpenCV, Python, PID Control, PLC, LiDAR, ROS 2, MATLAB, Gazebo, Embedded Systems

Methods: Autonomous navigation, Control loops, Sensor fusion

## PROFESSIONAL EXPERIENCE

**Robotics & Control Engineer | NovaPeak Engineering | Jul 2024 - Present**

- Reduced robotic path completion time by 40% through route and controller tuning
- Improved pick-position repeatability by 29% after calibration updates
- Validated 724 robot simulation scenarios with recorded sensor logs

## PROJECTS

**Vision-Guided Pick Cell | Embedded Systems, ROS 2, PLC, Gazebo**

- Integrated camera calibration, pick pose estimation and manipulator trajectories.
- Documented acceptance criteria and evaluated the design using control loops and sensor fusion.

**Autonomous Mobile Robot | PLC, MATLAB, PID Control, OpenCV**

- Built ROS 2 nodes for localization, obstacle avoidance and route execution.
- Documented acceptance criteria and evaluated the design using control loops and sensor fusion.

## EDUCATION

B.Tech in Mechatronics Engineering, Western Plains Institute of Technology, 2024
Relevant Methods: Autonomous navigation, Control loops, Sensor fusion

## CERTIFICATIONS

ROS 2 Navigation Workshop (2025)